PORTAL PURPOSE & OVERVIEW

Attendance Management Portal

One central place to manage cross-project team staffing, mark daily shift attendance, and download executive reports.

Why We Built It

Engineers work across multiple client projects and government sites. Management previously had no single, reliable view of who is absent without info, who is on leave with task cover, who arrived late, and who is falling short of the 9-hour shift. Attendance was isolated in separate per-project spreadsheets with no history and no trends.

The portal provides **one daily operational record** for every team filled by the TL/PM, with automatic checks, an executive daily summary for management, and long-term trend analysis.

1. Central User Directory

Loads employees automatically from company Keycloak SSO. Search anyone easily and move engineers between projects in 1 click.

2. Daily Shifts & Leaves

Mark project attendance quickly, track exact check-in and check-out times, WFH, half-days, and pre-fill approved multi-day leaves automatically.

3. Reports, Trends & Excel Downloads

See daily summaries, review attendance trends (7 to 365 days or custom dates) with instant engineer search, and export formatted Excel sheets in 1 click.

WHY WE BUILT THIS

Common Problems in Team Attendance

Managing teams across multiple projects using spreadsheets creates confusion and wasted time.

1. No Proper Tracking of Check-In & Check-Out Times

Engineers check in and out at different times without accurate time records. Without timestamps, project leads cannot track punctuality, late arrivals, or total hours worked properly.

2. Scattered Spreadsheets & Hidden Bench Pool

Project leads kept isolated files. Management could not easily see available bench engineers for new project staffing.

3. Repeating the Same Leave Entry Every Day

When an engineer took an approved 5-day leave, the manager had to open the sheet and type "Leave" every single morning.

4. Missing Long-Term Attendance Trends

Management had no simple way to spot repeated late arrivals or compare team punctuality over months or weekdays.

5. Confusion Over Client Site vs. Head Office Duty

When engineers visited the Head Office for sprint reviews or IT setups, no clear note was recorded, creating confusion in timesheets.

TEAM MANAGEMENT

https://pm.keenable.io/attendance/team

Loading Users from Keycloak & Assigning Projects

Centralized login via Keycloak SSO, live directory search, and moving engineers between teams in 1 click.

Automatic Keycloak User Sync

Users log in with their regular company Keycloak credentials. Their name, email, and role sync automatically with zero manual setup.

Fast Live Name & Email Search

Type any part of a name or email to immediately find available engineers from the bench pool or existing projects.

1-Click Project Reallocation

Admins can transfer an engineer to any target project with custom shift start times while keeping their past attendance records safe.

User / Email	Role	Current Status	Action Available	Target Project	Shift Start
Mohd Imran imran.mohd@fosteringlinux.com	Software Engineer	BENCH POOL	**Assign to Team**	**UPSC**	09:30
Priya Sharma priya.s@fosteringlinux.com	Frontend Engineer	NCLT	**Transfer & Reassign**	**UPSC**	10:00
Vikram Singh vikram.s@fosteringlinux.com	DevOps Engineer	BENCH POOL	**Assign to Team**	**Shiksha**	09:30
Arun Kumar arunk.pal@fosteringlinux.com	Tech Lead	UPSC	**Update Shift / Role**	**NSP**	09:30

DAILY OPERATIONS

https://pm.keenable.io/attendance/

Daily Project Attendance & Shift Operations

Simple daily grid for project leads to mark check-ins, check-outs, lateness, leaves, and handovers.

Project Tabs

Switch smoothly between project tabs (UPSC, NCLT, Shiksha, NSP). Each member is shown with their shift start time.

Clear Status Codes

P Present LT Late L Leave HD Half Day W WFH A Absent

Automatic Late Calculation

Compares check-in time against the engineer's shift start (e.g. 09:30 + 15m grace) and calculates late minutes in a dedicated column.

Engineer	Shift Start	Status	In Time	Out Time	Late (Mins)	Cover Handover	Daily Remarks / Context
Arun Kumar (Lead)	09:30	Present	09:25	18:35	—	—	Client site deployment
Mohd Imran	09:30	Present	09:55	18:30	+25m	—	Metro transit delay
Priya Sharma	10:00	Leave	—	—	—	Vikram Singh	Advance Casual Leave (Approved on email)
Vikram Singh	09:30	Present	09:20	18:40	—	—	Visited Head Office for internal review
Neha Verma	09:30	WFH	09:30	18:30	—	—	Approved remote duty (Sprint coding)
Suresh Patel	09:30	Half Day	09:30	14:00	—	Arun Kumar	First-half shift · Personal medical appointment
Amit Roy	09:30	Leave	—	—	—	Mohd Imran	Advance Sick Leave · Day 2 of 3

OPERATIONAL SUMMARY

https://pm.keenable.io/attendance/summary

Daily Summary & Excel Downloads

Instant daily numbers, task handover verification, and 1-click Excel sheets ready for management review.

TOTAL MEMBERS	PRESENT TODAY	ON LEAVE / WFH	LATE ARRIVALS
28	22	5	2

Clear Daily Highlights

- **Absent without info**: Shows unplanned absences needing follow-up.
- **On Leave & Handover**: Shows who is covering tasks + open OpenProject tickets.
- **Present Members Table**: Full list with shift start, in/out times, and remarks.

1-Click Excel Reports

- **Download Today's Sheet**: Instant formatted Excel file for today's entries.
- **Monthly Report**: Full monthly matrix + complete daily log sheet.
- **Email Summary Now**: Sends management an instant HTML email summary.

PUNCTUALITY TRENDS

https://pm.keenable.io/attendance/trends

Attendance Trends: 7 to 365 Days & Custom Dates

Interactive graphs, weekday lateness patterns, and scorecards covering short, medium, and multi-year ranges.

Extended Time Ranges

Filter by **7, 14, 30, 60, 90, 180 days (6 months), 365 days (1 year)**, or pick any **Custom Date Range** from past years.

🔍 Real-Time Engineer Search

Autofocus combobox (🔍 Search name or role...) filters individual trends instantly with keyboard nav (Enter / Esc) & 1-click clear.

Weekday Matrix & Regularity

Identifies recurring late days (e.g. *Monday 38%* vs *Friday 4%*) and grades attendance as **Regular**, **Minor**, or **Irregular**.

Engineer	Regularity Flag	Days Present	Late Days	Late %	Avg Late	Planned Leaves	Punctuality Trend
Arun Kumar	REGULAR	22 of 22	1	4%	8m	0	Steady
Mohd Imran	MINOR	20 of 22	3	15%	18m	2	Improving
Priya Sharma	REGULAR	18 of 22	0	0%	—	4	Steady

LEAVE AUTOMATION

Modal on https://pm.keenable.io/attendance/

Pre-Marking Approved Leaves in Advance

Enter pre-approved email leaves once; future dates are automatically marked while skipping weekends.

1. One-Time Setup

When leave is approved on email, the PM enters the date range (e.g. Oct 20 to Oct 24), leave type, and chooses a task cover person.

2. Skips Weekends Automatically

The system automatically skips Saturdays and Sundays. Only official working days are pre-marked as Leave (`L`).

3. Easy Early Return

If the engineer comes back to work early (e.g. Thursday), simply mark today as **Present**; remaining future leave days stay saved.

Visual Flow: Email Approval to Daily Roster

1. Approved on Email
Manager approves request

➔

2. Enter Date Range
PM enters span in modal

➔

3. Auto-Prepopulates
Weekdays marked as 'L'

➔

4. Ready Every Morning
No daily manual typing

LOCATION TRANSPARENCY

Remarks on / & /summary

Client Site vs. Head Office Duty Remarks

Recording off-site work locations clearly so internal project timesheets remain 100% accurate.

The Real-World Situation

Engineers usually work at client sites. When visiting Head Office for sprint reviews, internal meetings, or IT hardware setup, they remain **actively on duty** for their project.

How Project Leads Mark It

1. Mark status as **Present (P)** on the daily grid.
2. Enter check-in and check-out times.
3. Type the simple reason in **Remarks**: *"Visited Head Office for internal sprint review / IT hardware setup"*.

Where the Note Appears

- **My Attendance (`/me`)**: Engineer can see their location note.
- **Summary Dashboard (`/summary`)**: Visible in Present Members list.
- **Daily Email Digest**: Sent in executive daily summary email.
- **Excel Downloads**: Saved in the monthly log for timesheet records.

EMPLOYEE VIEW

https://pm.keenable.io/attendance/me

My Attendance: Self-Service Employee View

A clear monthly calendar where engineers can check their daily punch times, shifts, and leaves.

Personal Monthly Calendar

Review check-in times, check-out times, and total hours worked for any selected month.

Shift Time Transparency

Displays assigned shift start time (e.g. 09:30), grace minutes, and highlights any late minutes.

Leave & Remarks Check

Verify pre-approved leaves and confirm that Head Office duty notes were recorded properly.

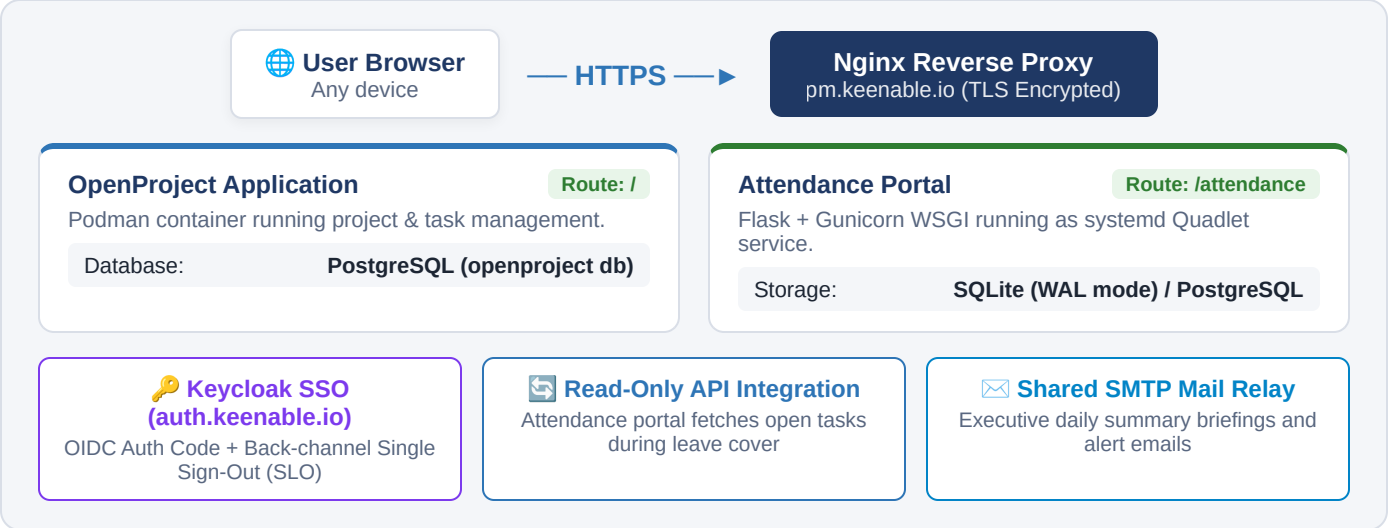
Date	Project	Status	In Time	Out Time	Worked Hours	Late By	Remarks
19-Oct Mon	UPSC	Present	09:25	18:35	9h 10m	—	Client site deployment
20-Oct Tue	UPSC	Present	09:20	18:40	9h 20m	—	Visited Head Office for internal review
21-Oct Wed	UPSC	Leave	—	—	—	—	Advance Casual Leave (Pre-approved on email)

ARCHITECTURE & TOPOLOGY

ONE DOMAIN · ONE SIGN-IN · SEPARATE DATA

Infrastructure Architecture & Dual-App Topology

Nginx routes / to OpenProject and /attendance to the Attendance Portal behind shared Keycloak SSO.



SECURITY & STACK

SMALL STACK · STANDARD SECURITY

Technology Stack & Security Architecture

Few moving parts, standard protocols, and enterprise-grade role-based access control.

Layer	Technology & Implementation
Application	Python 3.12, Flask 3.0, Gunicorn WSGI (--threads 8)
Pages & UI	Server-rendered Jinja2 templates, inline dynamic SVG charts
Database	SQLite 3 in container volume (WAL mode; PostgreSQL supported via DATABASE_URL)
Identity & SSO	Keycloak, OpenID Connect Authorization Code Flow (Authlib)
Reports	openpyxl for high-speed Excel export generation

Enterprise Security & Governance

• **Zero Password Storage**: Portal stores no credentials; authentication is 100% delegated to Keycloak.
• **Single Sign-Out (SLO)**: Signing out in either app immediately terminates session across OpenProject & Keycloak via back-channel webhook.
• **Dynamic 5-Minute Role Re-Check**: Keycloak roles are refreshed every 5 minutes. A revoked role or disabled account loses access within 5 minutes.
• **Role-Based Access Control (RBAC)**:
  – *Admin*: Full control over all projects, settings, and bench pool.
  – *TL / PM*: Entry, leave modal, and summary for assigned projects only.
  – *Engineer*: Self-service personal attendance view (/me) only.
• **CSRF & Session Security**: Anti-CSRF token on every POST, HTTPS-only secure cookies.
• **Configurable Settings**: Shift start (default 09:30), 15m grace, 9h shift length, and weekly offs are settings, not hardcoded.

DELIVERY STATUS

PRODUCTION STATUS & ROADMAP

Production Delivery Status & Roadmap

Clear inventory of live capabilities delivered in production vs planned next-phase enhancements.

Feature / Capability	Status	Implementation Details & Value
Projects, Teams & Daily Entry	LIVE	Live on pm.keenable.io/attendance/ with multi-project tabs, in/out timestamps, and lateness tracking.
Keycloak SSO & 5-Min Role Re-Check	LIVE	Central single sign-on with automatic back-channel Single Sign-Out (SLO) and 5-minute permission re-sync.
Keycloak User Directory & Bench Pool	LIVE	Instant directory search on /team, bench pool visibility, and 1-click project reassignment.
Advance Leave Modal (Skip Weekends)	LIVE	Multi-day leave pre-filling that skips Saturdays & Sundays with mandatory task cover validation.
WFH & Half-Day Status Tracking	LIVE	Dedicated status codes (W and HD) with timestamp logs and location remarks.
Extended Trends & Engineer Search	LIVE	7, 14, 30, 60, 90, 180, 365 days, custom date range, and instant engineer search bar with keyboard nav.
Excel Downloads & Email Briefing	LIVE	Single-day and monthly Excel workbooks + automated management briefing email dispatch.
Audit-Lock Past Dates for TL/PM	NEXT	Freeze past dates for TL/PM to ensure audit-proof history; corrections restricted to Admin override.
PostgreSQL Server Migration	NEXT	Already architected: set DATABASE_URL to transition from local SQLite to dedicated PostgreSQL.

LIVE DEMONSTRATION

FULL USER JOURNEY

Step-by-Step Live Demonstration Flow

A realistic end-to-end walkthrough showing how all roles use the portal in practice.

1. **Admin Searches Directory & Reassigns Bench Engineer** `/team`
   Admin searches for an engineer loaded from Keycloak and assigns them from Bench Pool to UPSC project with 09:30 shift start.

2. **PM Marks Daily Attendance & Head Office Remark** `/`
   PM selects UPSC project tab, records check-in/out times, views late minutes, and notes a Head Office visit in Remarks.

3. **PM Pre-Marks Pre-Approved Advance Leave** LEAVE MODAL
   PM opens Leave Management modal and enters an approved 4-day leave. Future weekdays are automatically pre-marked.

4. **PM Reviews Summary & Downloads Monthly Excel** `/summary` `/trends`
   PM checks daily numbers on /summary, inspects long-term regularity on /trends with instant engineer search, and downloads the monthly Excel report.

5. **Engineer Checks Personal Monthly Calendar** `/me`
   Engineer logs in using Keycloak to verify their shift start time, punch hours, approved leave days, and Head Office duty note.